NEET UG 2027: Master Biology Chapter Notes

Ecology, Biodiversity & Systematic Morphology of Plant Families

Prepared, Compiled & Authored by Satya Prakash · Strictly Calibrated to Rationalized NCERT & NMC Scope

1 Organisms & Populations

Population Growth Formulations: Exponential vs Logistic Model

1. Exponential Growth (Unlimited Resources / Geometric Progression):

$$\frac{dN}{dt} = (b-d)N = rN \implies N_t = N_0 e^{rt} \quad \textbf{(J-shaped curve)} \tag{1}$$

Where N = population size, b = per capita birth rate, d = per capita death rate, r = **intrinsic rate of natural increase**:

- Norway rat: $r = 0.015$ | Flour beetle: $r = 0.12$ | Human population in India (1981): $r = 0.0205$.

2. Logistic Growth / Verhulst-Pearl Model (Limited Resources / Realistic):

$$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right) \quad \textbf{(Sigmoid / S-shaped curve)} \tag{2}$$

Where K = **Carrying Capacity** (maximum population density the habitat can support indefinitely).

- Four phases: Lag phase → Acceleration / Log phase → Deceleration → Asymptote ($N = K$).
- When $N \to K$, environmental resistance $\left(\frac{K-N}{K}\right) \to 0$, growth rate drops to zero.

Interspecific Population Interactions Matrix

Interaction Type	Species A	Species B	NCERT Canonical Examples
Mutualism	+	+	Lichens (Algae + Fungi); Mycorrhizae (Fungi + Roots of higher plants); Fig tree and pollinator Fig Wasp; Mediterranean orchid *Ophrys* (**sexual deceit** by mimicking female bee *Colpa*).
Competition	−	−	Abingdon tortoise in Galapagos became extinct within a decade after goats were introduced (browsing efficiency); **Gause's Exclusion Principle** (inferior eliminated if resources limiting); **MacArthur's Warblers** (5 species coexisted via **resource partitioning** and foraging behavior).
Predation	+	−	Biological control: *Cactoblastis* moth controlled invasive prickly pear cactus in Australia (1920); Pisaster starfish removal caused extinction of > 10 invertebrate species; Monarch butterfly accumulates cardiac glycosides from weed in caterpillar stage.
Parasitism	+	−	Ectoparasites: Lice on humans, ticks on dogs, marine copepods on fishes, *Cuscuta* (dodder) on hedge plants; Endoparasites: Liver fluke (requires snail and fish hosts), Malarial parasite; **Brood Parasitism** (Cuckoo/Koel lays eggs in Crow's nest).
Commensalism	+	0	Epiphytic orchid on mango tree branch; Barnacles attached to back of whale; Cattle egret foraging alongside grazing cattle; Clownfish living among stinging tentacles of Sea anemone.
Amensalism	−	0	*Penicillium* secreting penicillin antibiotic inhibiting bacterial growth; Black walnut (*Juglans nigra*) secreting juglone inhibiting understory plants.

2 Ecosystem: Structure, Function & Ecological Succession

Productivity Equations & Decomposition Dynamics

1. Primary Productivity Formulations:

Gross Primary Productivity (GPP): Total rate of organic matter synthesis in photosynthesis.
Net Primary Productivity (NPP): GPP $-$ R (R = respiratory loss)
Global Annual Biosphere NPP: $\approx$ **170** billion tons (dry organic matter)
Oceans Productivity: **55** billion tons (Covers 70% but yields 32% NPP)

2. Stages of Decomposition: Detritus (dead leaves, bark, animal remains) undergoes five simultaneous steps:

1. **Fragmentation:** Breakdown of detritus into smaller particles by detritivores (e.g., Earthworm).
2. **Leaching:** Water-soluble inorganic nutrients percolate into soil horizon and precipitate as unavailable salts.
3. **Catabolism:** Bacterial and fungal enzymes degrade detritus into simpler inorganic substances.
4. **Humification:** Accumulation of dark, amorphous, colloidal **Humus** (highly resistant to microbial action, extremely slow decomposition, acts as reservoir of nutrients).
5. **Mineralization:** Slow release of inorganic nutrients by microbial degradation of humus.

Rate Factors: Decomposition is fast if detritus is rich in **nitrogen and water-soluble sugars**; slow if rich in **lignin and chitin**.

Energy Flow & Ecological Pyramids Architecture

- **10% Law of Lindeman:** Only **10**% of energy entering a trophic level is transferred to the next higher level; 90% is lost as metabolic heat.
- **Photosynthetically Active Radiation (PAR):** Out of total incident solar radiation, less than **50**% is PAR. Plants capture only **2 – 10**% of PAR (1 – 5% of total incident radiation).
- **Ecological Pyramids Comparison:**
  - **Pyramid of Numbers:** Mostly upright (grassland); **inverted** in tree ecosystem supporting thousands of parasitic insects.
  - **Pyramid of Biomass:** Upright on land (trees/grasses have vast biomass); **Inverted in Aquatic / Marine Ecosystems** (biomass of phytoplankton is drastically smaller than that of zooplankton and predatory fishes).
  - **Pyramid of Energy: Strictly and always upright**; can never be inverted because energy is lost as heat at each transfer (Second Law of Thermodynamics).
- **Succession Trajectory:** Both **Hydrarch** (water) and **Xerarch** (bare rock) successions converge toward identical moderate water conditions called **Mesic conditions**.

3 Biodiversity & Conservation

Species-Area Relationship (Alexander von Humboldt)

Alexander von Humboldt observed that within a region, species richness increases with increasing explored area, but only up to a limit:

$$S = CA^Z \implies \log \mathbf{S} = \log \mathbf{C} + \mathbf{Z} \log \mathbf{A} \tag{3}$$

Where S = Species richness, A = Area, C = Y-intercept, Z = **Slope of regression line**:

- For small regional areas (plants in Britain, birds in California): Z is strictly 0.1 **to** 0.2.
- For vast continents (frugivorous birds and mammals in tropical forests): Z is much steeper, 0.6 **to** 1.2 (typically 1.15).

Global Biodiversity Metrics & The "Evil Quartet" Causes of Loss

1. Biodiversity Estimates (Robert May): Global species total $\approx$ **7** million. More than **70% of all recorded species are animals**, and plants comprise $\leq 22\%$. Among animals, **insects make up** $> 70\%$ (7 out of 10 animals on Earth are insects). India has **2.4% of global land area** but accounts for **8.1% of global species diversity** (one of 12 mega-diversity countries).

2. The "Evil Quartet" (Major Drivers of Biodiversity Extinction):

1. **Habitat Loss & Fragmentation:** The single most catastrophic cause. Tropical rainforests once covered 14% of Earth's land, now reduced to $< 6\%$. Amazon rainforest is being cleared for soybean cultivation and cattle pasture.
2. **Over-Exploitation:** Excessive hunting led to extinction of *Steller's sea cow* and *Passenger pigeon*.
3. **Alien Species Invasions:**
   - Nile perch introduced into Lake Victoria caused extinction of > 200 endemic *Cichlid fish* species.
   - Invasive weeds: *Parthenium* (Carrot grass), *Lantana*, and *Eichhornia crassipes* (Water Hyacinth / "Terror of Bengal").
   - Illegal introduction of African catfish *Clarias gariepinus* threatens indigenous catfishes in Indian rivers.
4. **Co-Extinctions:** Obligate co-evolved relationships (e.g., host fish and its specific monogenean parasite; plant-pollinator mutualism).

3. In-Situ vs Ex-Situ Conservation:

- **In-Situ (On-site):** 14 Biosphere Reserves, 90 National Parks, 448 Wildlife Sanctuaries, Sacred Groves (Khasi and Jaintia Hills in Meghalaya, Aravalli Hills in Rajasthan, Western Ghats).
- **Ex-Situ (Off-site):** Zoological parks, Botanical gardens, Wildlife safari parks, Cryopreservation of gametes at $-196°$C in liquid nitrogen, Seed banks, Tissue culture.

4 Systematic Morphology of Plant Families (Prescribed NMC Syllabus)

Diagnostic Master Comparison of Prescribed Angiosperm Families

Family	Floral Formula	Androecium / Gynoecium	Diagnostic Traits & Important Economic Plants
Brassicaceae (Mustard)	$\oplus\ K_{2+2}\ C_4\ A_{2+4}\underline{G}_{(2)}$	**Tetradynamous** (2 outer short, 4 inner long); Bicarpellary, unilocular with **false septum (Replum)**.	Cruciform corolla (clawed petals); Siliqua fruit. *Brassica campestris* (mustard), *Raphanus sativus* (radish).
Fabaceae (Legume)	$\%\ K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\underline{G}_1$	**Diadelphous** (9 fused +1 free); Monocarpellary, marginal placentation.	**Vexillary aestivation** (1 standard, 2 wings, 2 keel); Root nodules with *Rhizobium*. Pulses (Gram, Arhar), Oil (Soyabean, Groundnut), Dye (*Indigofera*), Fodder (*Sesbania*), Medicine (*Muliathi*).
Solanaceae (Potato)	$\oplus\ K_{(5)}\ \overbrace{C_{(5)}\ A_5}\ \underline{G}_{(2)}$	Epipetalous stamens; Bicarpellary, syncarpous, **oblique ovary**, **swollen axile placenta**.	Persistent calyx; Berry or capsule. Food (Potato, Tomato, Brinjal), Spice (Chilli), Medicine (*Belladonna*, *Ashwagandha*), Fumigatory (Tobacco), Ornamental (*Petunia*).
Liliaceae (Lily)	$\oplus\ \overbrace{P_{(3+3)}\ A_{3+3}}\ \underline{G}_{(3)}$	Epiphyllous stamens; Tricarpellary, syncarpous, trilocular with axile placentation.	Perianth of 6 tepals in two whorls; Capsule/berry. Ornamental (Tulip, *Gloriosa*), Medicine (*Aloe*), Vegetable (*Asparagus*), Colchicine (*Colchicum autumnale*).
Malvaceae (Cotton)	$\oplus\ K_{(5)}\ C_5\ A_{(\infty)}\underline{G}_{(5)}$	**Monadelphous** staminal tube; **Monothecous, reniform anthers**.	**Epicalyx** present; Mucilaginous sap; Twisted aestivation. Cotton (*Gossypium*), Lady's finger (*Abelmoschus*), China rose (*Hibiscus*).
Compositae (Sunflower)	Ray: $\%\ K_{pappus}\ C_{(5)}\ A_0\underline{G}_{(2)}$ Disc: $\oplus\ K_{pappus}\ \overbrace{C_{(5)}\ A_{(5)}}\ \underline{G}_{(2)}$	**Syngenesious** (anthers fused, filaments free); Bicarpellary, unilocular, **basal placentation**.	**Capitulum inflorescence**; Calyx modified into hairy **Pappus**; Cypsela fruit. Sunflower (*Helianthus*), Marigold (*Tagetes*), *Chrysanthemum*.
Gramineae (Poaceae - Grass)	$\%\ P_2\ A_3$ or $A_{3+3}\underline{G}_1$	Versatile anthers; Monocarpellary, unilocular, basal placentation with **feathery bifid stigma**.	**Spikelet inflorescence**; Perianth represented by 2 scale-like **Lodicules**; Caryopsis fruit. Cereals (Wheat, Rice, Maize, Sugarcane, Bamboo).